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Candidate Rotation Periods for 42 Main-belt Asteroids from Gaia DR3 Sparse Photometry

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ABSTRACT

I searched Gaia DR3 sparse photometry for asteroid rotation periods with a periodogram that combines season offsets, a linear phase term and a second-order Fourier series. Among the numbered asteroids that Gaia observed most often, I analysed the 300 highest-ranked with a known period (validation set) and the 298 highest-ranked without one (target set). On 337 confident results that can be compared with published periods, 80% agree within 3%. Accuracy is 81–89% for periods below 50 h and 36% above it, where the method usually returns half of the true period. I report 42 asteroids that had no rotation period in LCDB, JPL SBDB, DAMIT, about 180 VizieR tables, or the Minor Planet Bulletin 2023–2026, with candidate periods and a reliability flag, as targets for confirmation.

DATA AND METHOD

I ranked the numbered asteroids by the number of Gaia DR3 Solar-System-object observations (Tanga et al. 2023). From the top of this list I took the first 300 that have a rotation period of 2–100 h in JPL SBDB, which follows the October 2023 revision of the LCDB (Warner et al. 2009), as the validation set. I took the first 298 without a JPL period as the target set. Every selected asteroid has at least 443 Gaia observations. Gaia photometry was averaged per transit, with a 0.03 mag error floor. For every transit, JPL Horizons gave the heliocentric distance r, the distance Δ from Gaia and the phase angle α, with Gaia as the observer. Times were corrected for light time.

The reduced magnitude $G - 5\log_{10}(r\Delta)$ was fit by weighted least squares with one offset per observing season, a linear phase term and a second-order Fourier series in rotational phase. I scanned 2–100 h on a frequency grid with a step of $0.1/T$, where T is the time span. Because most asteroid lightcurves are double-peaked, I adopted the doubled period whenever doubling raised χ^2 by less than a factor of 1.5. This factor was chosen by two-fold cross-validation on the validation set. The confidence score combines the χ^2 gain over a no-rotation model with the separation of the best minimum from the next one outside the $\times 1$, $\times 2$ and $\times \frac{1}{2}$ aliases. The confidence threshold was set on the validation set, at the score where precision reaches 80%.

VALIDATION

The cross-validated recovery rate on the validation set is 67.7%. Above the threshold, I compared every result whose period is known (Figure 1):

- the validation set: 172 of 214 correct. This figure is in-sample, because the same set fixed the threshold, and is therefore optimistic;
- target-set asteroids that Ďurech & Hanuš (2023) solved from the same Gaia data with a different method: 81 of 97 agree. This tests the method, not the data;
- 26 of my initial candidates that later appeared in Cellino et al. (2024), MacLennan et al. (2026), McNeill et al. (2023) or Liberato et al. (2024): 17 of 26 agree.

In total, 269 of 337 (80%) agree.

There are two failure modes. Above 50 h the method usually returns half of the true period (10 of 28 correct). Slow rotators with periods beyond 100 h, such as 214, 818 and about 4,600 h, receive spurious short periods.

RESULTS

The 42 candidates are all main-belt asteroids. In October 2026 they were absent from:

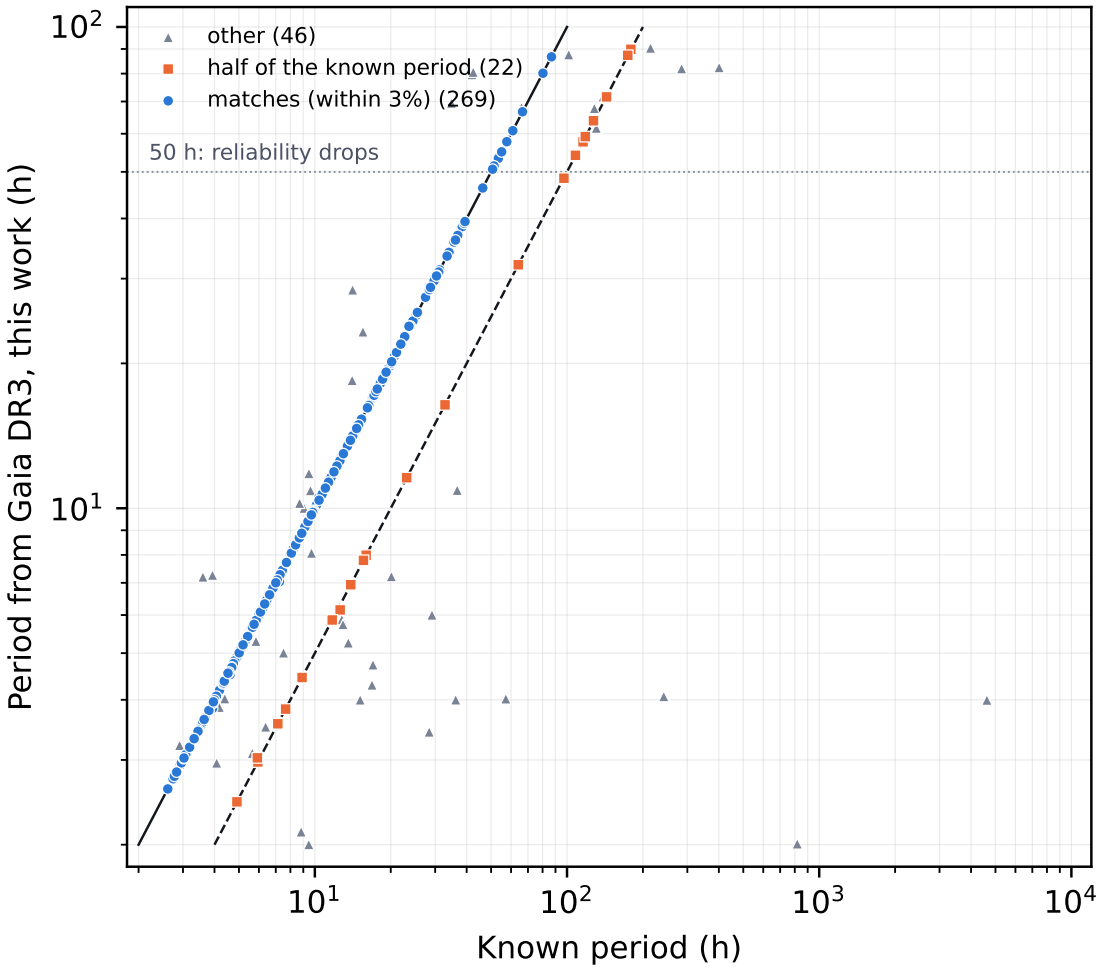


Figure 1. Period from Gaia DR3 (this work) against the published period for 337 confident results. The solid line marks agreement and the dashed line half of the known period. Above 50 h the method usually finds half of the true period.

- JPL SBDB and LCDB V4.0;
- Ďurech & Hanuš (2023);
- DAMIT (Ďurech et al. 2010);
- about 180 VizieR tables (5.1 million rows) that hold asteroid rotation periods;
- the full text of the Minor Planet Bulletin, volumes 50–53.

For the 34 candidates below 50 h I expect about 83% to be correct. For the 8 above 50 h, the true period is probably twice the value or longer. Periods, alias periods, amplitudes, amounts of data and flags are available at https://github.com/olegtropinin/gaia-asteroid-periods and archived on Zenodo (Tropynin 2026), together with the code and the validation tables. I offer the list as targets for dense-lightcurve confirmation.

This work used data from the ESA mission Gaia, processed by the Gaia DPAC; JPL Horizons and the JPL Small-Body Database; and VizieR (CDS, Strasbourg). The analysis was designed, coded and checked with the help of an AI assistant (Claude, Anthropic) under the author's direction.

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